

Day 2 Problem Set Solutions

Methods Camp | University of Texas at Austin

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Day 2 Problem Set

This document contains solutions to all exercises from the Day 2 problem set. To reinforce your understanding, please try your best to complete these exercises independently before referring to the solutions.

Exercise 1: Random Variables & Expectation

A survey asks respondents how many times they attended religious services last month. Suppose the distribution is:

Times attended (x)	0	1	2	4
$P(X = x)$	0.4	0.3	0.2	0.1

1.1 Is X discrete or continuous? Why?

1.2 Compute $E[X]$ by hand.

1.3 In your own words, what does $E[X]$ tell us?

1.4 If attendance is recoded as $Y = 2X + 1$, what is $E[Y]$? (Use the linearity property.)

Exercise 2: Distributions

2.1 For each scenario below, name the most appropriate distribution (Bernoulli, Binomial, Poisson, or Normal) and briefly justify your choice:

- Whether a single respondent owns a home (yes/no)
- The number of home-owners among 200 randomly sampled respondents

- c. The number of hate-crime incidents reported in a city per month
- d. Adult height in a large population

2.2 For scenario (b), if $p = 0.65$, what is $E[X]$?

Exercise 3: Regression

Suppose a researcher wants to examine occupational prestige score (Y) as a linear function of parents' years of schooling (X) across all U.S. families.

- 3.1 Write the regression equation using index notation. What does the subscript i represent?
- 3.2 Interpret the slope coefficient.
- 3.3 Identify the estimand, a plausible estimator, and an example of an estimate.
- 3.4 Name one plausible factor that could be included in the error term, ε_i .

Solutions

Exercise 1: Random Variables & Expectation

1.1 X is **discrete**: it only takes on a countable set of values (0, 1, 2, 4), not any value in a continuous interval.

$$1.2 E[X] = 0(0.4) + 1(0.3) + 2(0.2) + 4(0.1) = 0 + 0.3 + 0.4 + 0.4 = 1.1$$

1.3 $E[X] = 1.1$ is the long-run average across many respondents — if we surveyed thousands of people and averaged their attendance, we'd land close to 1.1.

$$1.4 \text{ Using linearity, } E[aX+b] = aE[X]+b: \quad E[Y] = E[2X+1] = 2E[X]+1 = 2(1.1)+1 = 3.2$$

Exercise 2: Distributions

2.1 (a) Bernoulli — a single yes/no trial for one respondent. (b) Binomial — the count of “successes” (home-owners) across $n = 200$ independent Bernoulli trials, each with the same probability p . (c) Poisson — a count of relatively rare events occurring over a fixed interval (a month), without a natural fixed number of “trials.” (d) Normal — a continuous measurement that tends to cluster symmetrically around an average, the classic bell-curve shape.

$$2.2 \text{ Using } E[X] = np \text{ for a Binomial: } E[X] = 200 \times 0.65 = 130$$

Exercise 3: Regression

3.1 $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$. The subscript i identifies a particular adult child or family.

3.2 β_1 represents the expected difference in the child's occupational prestige score associated with one additional year of parental schooling.

3.3 **Estimand**: the population slope β_1 ; **Estimator**: ordinary least squares (OLS); **Estimate**: the value obtained from the sample, such as $\hat{\beta}_1 = 1.5$.

3.4 Parental income, neighborhood conditions, or the local labor market.